

MITRC
**MODERN INSTITUTE OF TECHNOLOGY AND
RESEARCH CENTER**

NAME OF FACULTY: PRADEEP KUMAR
SUBJECT/ SUB-CODE: SMF/ 6AD4-03
SEMESTER: VI
BRANCH: AI&DS
DATE OF SUBMISSION: 16-03-2026

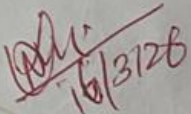
SESSION- 2025-2026
BATCH- C

Topic Covered: 70 Important Question and Answers

PART-A (40 Questions)
PART-B (15 Questions)
PART-C (15 Questions)


FACULTY


HOD


16/3/26
DEAN ACADEMIC

PART-A

Q1. What is Time Series Data?

A time series is a sequence of observations collected at regular time intervals such as daily temperature or monthly sales. It helps analyze patterns and forecast future values. Mathematically, a time series can be written as

$$Y_t = f(t)$$

where Y_t represents the observation at time t . Time series analysis identifies components like trend, seasonality, cyclic variation, and random noise to understand the structure of data and build forecasting models.

Q2. What are the moments of a distribution in time series?

Moments describe the statistical characteristics of a distribution. The first moment is the mean

$$\mu = E(X)$$

The second moment about the mean is the variance

$$\sigma^2 = E[(X - \mu)^2]$$

Higher moments include skewness and kurtosis. Skewness measures asymmetry while kurtosis measures peakedness. These moments help describe the shape and variability of time series data and are important for statistical modeling and forecasting accuracy.

Q3. What is stationarity in time series analysis?

A time series is stationary if its statistical properties remain constant over time. The conditions for stationarity include constant mean, constant variance, and constant autocovariance. These are expressed as

$$\begin{aligned} E(Y_t) &= \mu \\ \text{Var}(Y_t) &= \sigma^2 \\ \text{Cov}(Y_t, Y_{t-k}) &= \gamma_k \end{aligned}$$

where k is the lag. Stationarity is required for many models like ARMA because stable statistical properties allow reliable forecasting and parameter estimation.

Q4. What is autocorrelation in time series?

Autocorrelation measures the relationship between observations separated by time lag. It shows how current values depend on previous values. The autocorrelation coefficient at lag k is

$$\rho_k = \frac{\text{Cov}(Y_t, Y_{t-k})}{\text{Var}(Y_t)}$$

Autocorrelation values range from -1 to $+1$. A high positive value indicates strong similarity between observations, while a negative value indicates opposite movement. It is useful in identifying patterns and selecting appropriate time series models.

Q5. What is heteroscedasticity in time series?

Heteroscedasticity occurs when the variance of the time series changes over time. In such cases the variance is not constant and can be expressed as

$$\text{Var}(Y_t) = \sigma_t^2$$

This situation often appears in financial data where periods of high volatility alternate with low volatility. Heteroscedasticity violates the assumption of constant variance used in many statistical models, so special models such as ARCH and GARCH are often applied.

Q6. What is normality in time series data?

Normality means the data follow a normal distribution. Many statistical methods assume that the residuals of a model follow a normal distribution. The probability density function of a normal distribution is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where μ is the mean and σ is the standard deviation. Normality helps in making statistical inference, hypothesis testing, and confidence interval estimation in forecasting models.

Q7. What are the components of a time series?

A time series usually consists of four main components: trend, seasonal variation, cyclical variation, and irregular variation. These components can be combined using the additive model

$$Y_t = T_t + S_t + C_t + I_t$$

or the multiplicative model

$$Y_t = T_t \times S_t \times C_t \times I_t$$

where T_t represents trend, S_t seasonal component, C_t cyclical movement, and I_t irregular fluctuations.

Q8. What is forecasting in time series?

Forecasting is the process of predicting future values of a time series based on past observations. A general forecasting model can be written as

$$\hat{Y}_{t+h} = f(Y_t, Y_{t-1}, Y_{t-2}, \dots)$$

where h is the forecast horizon. Forecasting is widely used in business, economics, weather prediction, and finance to assist decision making and planning activities.

Q9. What are different types of data used in forecasting?

Forecasting commonly uses cross-sectional data, time series data, and panel data. Time series data involve observations over time such as

$$Y_1, Y_2, Y_3, \dots, Y_t$$

Cross-sectional data contain observations at a single point in time across different units. Panel data combine both time series and cross-sectional dimensions and can be represented as

$$Y_{it}$$

where i denotes the individual and t represents time.

Q10. What is the forecasting process?

The forecasting process involves identifying the problem, collecting data, selecting a model, estimating parameters, validating the model, and producing forecasts. A simple forecasting equation is

$$\hat{Y}_{t+1} = Y_t + e_t$$

where e_t is the random error term. Proper evaluation using accuracy measures such as Mean Squared Error ensures that the selected model provides reliable future predictions.

Q11. What are graphical displays in time series analysis?

Graphical displays help visualize patterns and trends in time series data. The most common plot is the time series plot where values are plotted against time. If t represents time and Y_t the observation, the plot shows points (t, Y_t) . Graphical displays allow analysts to detect trends, seasonal patterns, and outliers, which helps in selecting appropriate forecasting models.

Q12. What is a time series plot?

A time series plot represents observations of a variable over time on a graph. The horizontal axis shows time while the vertical axis shows the observed value Y_t . The data points follow the sequence

$$Y_1, Y_2, Y_3, \dots, Y_t$$

Such plots help visualize trends, cycles, and seasonal variations, making it easier to understand the underlying structure of the data before applying forecasting models.

Q13. What is smoothing in time series data?

Smoothing is a technique used to remove short-term fluctuations and highlight long-term trends. One common method is the moving average

$$MA_t = \frac{Y_t + Y_{t-1} + Y_{t-2} + \dots + Y_{t-n+1}}{n}$$

where n is the window size. Smoothing helps reduce noise in the data and makes it easier to observe the overall pattern or trend.

Q14. What is numerical description of time series data?

Numerical description summarizes time series data using statistical measures such as mean, variance, and standard deviation. The mean is calculated as

$$\bar{Y} = \frac{1}{n} \sum_{t=1}^n Y_t$$

Variance measures dispersion

$$\sigma^2 = \frac{1}{n} \sum_{t=1}^n (Y_t - \bar{Y})^2$$

These measures help understand central tendency and variability of the time series data.

Q15. What are data transformations in time series analysis?

Data transformations are used to stabilize variance and make the data more suitable for modeling. A common transformation is the logarithmic transformation

$$Z_t = \log(Y_t)$$

Another is differencing

$$Z_t = Y_t - Y_{t-1}$$

These transformations help remove trends or heteroscedasticity and convert non-stationary data into stationary data for better model performance.

Q16. What is the random walk model?

A random walk model assumes that the current value depends on the previous value plus a random error. It is expressed as

$$Y_t = Y_{t-1} + \varepsilon_t$$

where ε_t is white noise. This model is commonly used in financial time series such as stock prices where future movements are unpredictable and follow random fluctuations.

Q17. What is a unit root process?

A unit root process is a non-stationary time series where shocks have permanent effects. It can be represented as

$$Y_t = Y_{t-1} + \varepsilon_t$$

If the coefficient of Y_{t-1} equals 1, the process has a unit root. Unit root tests such as the Augmented Dickey–Fuller test are used to determine whether differencing is required to make the series stationary.

Q18. What is a drift model in time series?

A drift model extends the random walk model by including a constant drift term that represents average growth. The model is

$$Y_t = c + Y_{t-1} + \varepsilon_t$$

where c is the drift parameter. The drift term introduces a systematic upward or downward trend in the series over time.

Q19. What is an autoregressive (AR) model?

An autoregressive model expresses the current value of a time series as a linear combination of its past values. An AR(p) model is written as

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + \varepsilon_t$$

where p is the order of the model. AR models capture persistence in data and are widely used in forecasting.

Q20. What is the moving average (MA) model?

A moving average model represents a time series as a function of past error terms. An MA(q) model is expressed as

$$Y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

where q is the order of the model. MA models are useful when the current value is influenced by past random shocks.

Q21. What is the ARMA model?

The ARMA model combines autoregressive and moving average components. It is suitable for stationary time series. The ARMA(p, q) model is

$$Y_t = c + \sum_{i=1}^p \phi_i Y_{t-i} + \varepsilon_t + \sum_{j=1}^q \theta_j \varepsilon_{t-j}$$

where p is the AR order and q is the MA order. ARMA models are widely used for modeling short memory stationary processes.

Q22. What is the ARIMA model?

ARIMA stands for Autoregressive Integrated Moving Average. It is used for non-stationary time series data. The model is written as ARIMA(p, d, q) where d represents the number of differencing operations. Differencing is

$$\nabla Y_t = Y_t - Y_{t-1}$$

After differencing, the resulting stationary series can be modeled using ARMA methods for forecasting.

Q23. What is seasonal ARIMA?

Seasonal ARIMA extends the ARIMA model to handle seasonal patterns in time series. It is written as

$$ARIMA(p, d, q)(P, D, Q)_s$$

where s is the seasonal period. Seasonal differencing is

$$Y_t - Y_{t-s}$$

This model is useful for monthly or quarterly data where seasonal patterns repeat regularly.

Q24. What is partial autocorrelation?

Partial autocorrelation measures the correlation between Y_t and Y_{t-k} after removing the effect of intermediate lags. It helps identify the order of autoregressive models. The partial autocorrelation function (PACF) is used to determine the appropriate lag length p in AR models.

Q25. What is the Vector Autoregressive (VAR) model?

VAR models are used for multivariate time series where multiple variables influence each other. A VAR(1) model can be written as

$$Y_t = A_1 Y_{t-1} + \varepsilon_t$$

where Y_t is a vector of variables and A_1 is a coefficient matrix. VAR models capture dynamic relationships among multiple time series variables.

Q26. What is the Impulse Response Function (IRF)?

The impulse response function measures the effect of a shock to one variable on another variable over time in a VAR model. If a shock ε_t occurs at time t , the response of the system at future periods is calculated using the dynamic structure of the VAR equations.

Q27. What is cointegration in time series?

Cointegration occurs when two or more non-stationary time series move together in the long run. If X_t and Y_t are non-stationary but a linear combination

$$Z_t = Y_t - \beta X_t$$

is stationary, then the variables are cointegrated. Cointegration indicates a long-term equilibrium relationship between variables.

Q28. What is an error correction model (ECM)?

An error correction model describes the short-run dynamics while maintaining long-run equilibrium between variables. It is expressed as

$$\Delta Y_t = \alpha + \beta \Delta X_t + \lambda(Y_{t-1} - \beta X_{t-1}) + \varepsilon_t$$

The term $(Y_{t-1} - \beta X_{t-1})$ represents the error correction component that adjusts deviations from equilibrium.

Q29. What are model selection criteria?

Model selection criteria help choose the best forecasting model. Two common criteria are Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC).

$$\begin{aligned} AIC &= -2\ln(L) + 2k \\ BIC &= -2\ln(L) + k\ln(n) \end{aligned}$$

where L is likelihood, k number of parameters, and n sample size. Lower values indicate better models.

Q30. What is time series regression?

Time series regression models the relationship between a dependent variable and one or more explanatory variables over time. The linear regression model is

$$Y_t = \beta_0 + \beta_1 X_{1t} + \beta_2 X_{2t} + \varepsilon_t$$

It helps analyze how external variables influence the behavior of a time series.

Q31. What is least squares estimation?

Least squares estimation is used to estimate regression parameters by minimizing the sum of squared errors. The objective function is

$$S = \sum_{t=1}^n (Y_t - \hat{Y}_t)^2$$

Minimizing this function provides estimates of regression coefficients that best fit the observed data.

Q32. What is prediction in regression models?

Prediction involves estimating future observations based on the fitted regression model. The predicted value is

$$\hat{Y}_{t+1} = \beta_0 + \beta_1 X_{t+1}$$

Prediction intervals are also calculated to measure uncertainty around forecasts.

Q33. What is model adequacy checking?

Model adequacy checking verifies whether the fitted model properly explains the data. Residuals should behave like white noise. The residual is

$$e_t = Y_t - \hat{Y}_t$$

Residual analysis helps detect model misspecification, autocorrelation, or heteroscedasticity.

Q34. What is exponential smoothing?

Exponential smoothing is a forecasting technique that assigns exponentially decreasing weights to past observations. The simple exponential smoothing formula is

$$\hat{Y}_{t+1} = \alpha Y_t + (1 - \alpha) \hat{Y}_t$$

where $0 < \alpha < 1$ is the smoothing constant.

Q35. What is first-order exponential smoothing?

First-order exponential smoothing is used for time series without trend or seasonality. The formula is

$$S_t = \alpha Y_t + (1 - \alpha) S_{t-1}$$

where S_t is the smoothed value and α is the smoothing parameter.

Q36. What is second-order exponential smoothing?

Second-order exponential smoothing is used when the time series contains a trend. It introduces two smoothing equations:

$$\begin{aligned} S_t &= \alpha Y_t + (1 - \alpha) S_{t-1} \\ S'_t &= \alpha S_t + (1 - \alpha) S'_{t-1} \end{aligned}$$

This helps estimate both level and trend components.

Q37. What are multivariate time series models?

Multivariate time series models analyze multiple time-dependent variables simultaneously. A general representation is

$$Y_t = A_1 Y_{t-1} + A_2 Y_{t-2} + \varepsilon_t$$

where Y_t is a vector of variables. These models capture interactions among variables over time.

Q38. What are panel data models?

Panel data models analyze data that contain both cross-sectional and time dimensions. A typical panel model is

$$Y_{it} = \alpha + \beta X_{it} + u_{it}$$

where i represents individuals and t represents time. Panel models allow more informative data analysis.

Q39. What is the fixed-effect model?

The fixed-effect model assumes that individual-specific effects are constant over time. It is expressed as

$$Y_{it} = \alpha_i + \beta X_{it} + u_{it}$$

where α_i represents the individual effect. This model controls for unobserved heterogeneity.

Q40. What is Principal Component Analysis in forecasting?

Principal Component Analysis (PCA) reduces the dimensionality of data while preserving variance. The transformation is

$$Z = WX$$

where W is the eigenvector matrix. PCA is useful in forecasting when dealing with large numbers of correlated variables because it extracts the most important components for analysis.

PART-B

Q1. Calculate the mean and variance of the time series data: 4, 6, 8, 10, 12.

Mean is calculated using

$$\begin{aligned}\bar{Y} &= \frac{1}{n} \sum_{t=1}^n Y_t \\ \bar{Y} &= \frac{4 + 6 + 8 + 10 + 12}{5} = 8\end{aligned}$$

Variance is

$$\begin{aligned}\sigma^2 &= \frac{1}{n} \sum (Y_t - \bar{Y})^2 \\ &= \frac{(4 - 8)^2 + (6 - 8)^2 + (8 - 8)^2 + (10 - 8)^2 + (12 - 8)^2}{5} \\ &= \frac{16 + 4 + 0 + 4 + 16}{5} = 8\end{aligned}$$

Thus the **mean = 8** and **variance = 8**.

Q2. Find the first difference of the time series: 12, 15, 18, 22, 25.

First differencing is used to make a series stationary.

$$\begin{aligned}\Delta Y_t &= Y_t - Y_{t-1} \\ 15 - 12 &= 3 \\ 18 - 15 &= 3 \\ 22 - 18 &= 4\end{aligned}$$

$$25 - 22 = 3$$

Therefore the **differenced series = 3, 3, 4, 3.**

Differencing removes trend and helps in applying ARIMA models.

Q3. Calculate a 3-period moving average for the data: 10, 12, 14, 16, 18.

Moving average formula is

$$MA = \frac{Y_t + Y_{t-1} + Y_{t-2}}{3}$$

First MA

$$(10 + 12 + 14)/3 = 12$$

Second MA

$$(12 + 14 + 16)/3 = 14$$

Third MA

$$(14 + 16 + 18)/3 = 16$$

Thus the **3-period moving averages are 12, 14, 16** which smooth short-term fluctuations and highlight trends.

Q4. Using the random walk model, find Y_5 if $Y_4 = 20$ and error $\varepsilon_5 = 3$.

Random walk model:

$$Y_t = Y_{t-1} + \varepsilon_t$$

Substitute values:

$$\begin{aligned} Y_5 &= 20 + 3 \\ Y_5 &= 23 \end{aligned}$$

Thus the predicted value of the series at time 5 is **23**. Random walk models assume that future values depend only on the previous value plus random error.

Q5. Forecast the next value using AR(1) model where $Y_t = 0.7Y_{t-1}$ and $Y_{t-1} = 10$.

Autoregressive model:

$$Y_t = \phi_1 Y_{t-1}$$

Substitute values:

$$\begin{aligned} Y_t &= 0.7 \times 10 \\ Y_t &= 7 \end{aligned}$$

Hence the forecasted value is **7**. AR models use past observations to predict future values.

Q6. Calculate simple exponential smoothing forecast if $Y_t = 50$, previous forecast $F_t = 45$, and $\alpha = 0.3$.

Formula:

$$\begin{aligned} F_{t+1} &= \alpha Y_t + (1 - \alpha) F_t \\ F_{t+1} &= 0.3(50) + 0.7(45) \\ &= 15 + 31.5 \\ &= 46.5 \end{aligned}$$

Therefore the **forecast for next period = 46.5**.

Q7. Calculate covariance between two series X and Y where X = 2,4,6 and Y = 3,5,7.

Mean of X

$$\bar{X} = 4$$

Mean of Y

$$\bar{Y} = 5$$

Covariance formula

$$\begin{aligned} Cov(X, Y) &= \frac{1}{n} \sum (X - \bar{X})(Y - \bar{Y}) \\ &= \frac{(2 - 4)(3 - 5) + (4 - 4)(5 - 5) + (6 - 4)(7 - 5)}{3} \\ &= \frac{4 + 0 + 4}{3} \\ &= 2.67 \end{aligned}$$

Thus covariance ≈ 2.67 .

Q8. Compute Mean Squared Error (MSE) if actual values are 20, 25, 30 and forecasts are 18, 27, 29.

Error:

$$e_t = Y_t - \hat{Y}_t$$

Errors:

2, -2, 1

Square errors:

4, 4, 1

MSE formula:

$$\begin{aligned} MSE &= \frac{\sum e_t^2}{n} \\ &= \frac{4 + 4 + 1}{3} \\ &= 3 \end{aligned}$$

Thus the **Mean Squared Error = 3**.

Q9. Find autocorrelation at lag 1 for data: 3, 5, 7.

Mean

$$\bar{Y} = \frac{3 + 5 + 7}{3} = 5$$

Autocorrelation formula:

$$\rho_1 = \frac{\sum(Y_t - \bar{Y})(Y_{t-1} - \bar{Y})}{\sum(Y_t - \bar{Y})^2}$$

Numerator

$$(-2)(0) + (0)(2) = 0$$

Denominator

$$4 + 0 + 4 = 8$$

$$\rho_1 = 0/8 = 0$$

Thus autocorrelation = **0**.

Q10. Forecast using ARIMA differencing for series 10, 14, 18, 22.

First difference:

$$14 - 10 = 4$$

$$18 - 14 = 4$$

$$22 - 18 = 4$$

Trend is constant.

Next forecast

$$22 + 4 = 26$$

Thus predicted value = **26**.

Q11. Explain the Autoregressive (AR) model in detail.

The Autoregressive model describes a time series where the current value depends on its previous values. It assumes that past observations influence the present observation. The general AR(p) model is

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + \varepsilon_t$$

where p is the order of the model and ε_t is white noise. AR models are useful for forecasting stationary time series and identifying persistence in data.

Q12. Explain the Moving Average (MA) model.

The Moving Average model expresses a time series as a function of past error terms rather than past observations. The MA(q) model is written as

$$Y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \cdots + \theta_q \varepsilon_{t-q}$$

where q is the order of the model. MA models capture short-term random shocks affecting the series and are often combined with AR models to form ARMA models.

Q13. Explain the ARIMA model used in forecasting.

ARIMA stands for Autoregressive Integrated Moving Average. It is widely used for forecasting non-stationary time series data. The model is denoted as ARIMA(p, d, q) where p is the AR order, d is the number of differences, and q is the MA order. Differencing is performed using

$$\nabla Y_t = Y_t - Y_{t-1}$$

After differencing, the stationary series is modeled using ARMA methods for forecasting future values.

Q14. Explain exponential smoothing in time series forecasting.

Exponential smoothing is a forecasting technique that assigns exponentially decreasing weights to past observations. Recent observations receive higher importance than older ones. The basic formula is

$$F_{t+1} = \alpha Y_t + (1 - \alpha)F_t$$

where α is the smoothing constant between 0 and 1. Exponential smoothing is widely used in business forecasting because it is simple and effective for short-term predictions.

Q15. Explain the Vector Autoregressive (VAR) model.

The Vector Autoregressive model is used for multivariate time series where several variables influence each other. In VAR models each variable depends on its own past values and the past values of other variables. A VAR(1) model can be written as

$$Y_t = A_1 Y_{t-1} + \varepsilon_t$$

where Y_t is a vector of variables and A_1 is the coefficient matrix. VAR models are widely used in economics and financial forecasting.

PART-C

Q1. Compute the mean, variance and standard deviation for the time series data: 5, 7, 9, 11, 13, 15.

Mean formula

$$\bar{Y} = \frac{1}{n} \sum_{t=1}^n Y_t$$

$$\bar{Y} = \frac{5 + 7 + 9 + 11 + 13 + 15}{6} = 10$$

Variance

$$\sigma^2 = \frac{\sum (Y_t - \bar{Y})^2}{n}$$

$$= \frac{25 + 9 + 1 + 1 + 9 + 25}{6} = 11.67$$

Standard deviation

$$\sigma = \sqrt{11.67} = 3.41$$

Thus **Mean = 10, Variance $\approx$ 11.67, Standard deviation $\approx$ 3.41**. These measures describe central tendency and dispersion in the time series dataset.

Q2. Calculate a 4-period moving average for the data: 12, 15, 18, 20, 24, 28, 30.

Moving average formula

$$MA = \frac{Y_t + Y_{t-1} + Y_{t-2} + Y_{t-3}}{4}$$

First MA

$$(12 + 15 + 18 + 20)/4 = 16.25$$

Second MA

$$(15 + 18 + 20 + 24)/4 = 19.25$$

Third MA

$$(18 + 20 + 24 + 28)/4 = 22.5$$

Fourth MA

$$(20 + 24 + 28 + 30)/4 = 25.5$$

Thus the **4-period moving averages are 16.25, 19.25, 22.5, 25.5**, indicating an increasing trend.

Q3. Apply first and second differencing to the series: 8, 13, 19, 26, 34.

First difference

$$\begin{aligned}\Delta Y_t &= Y_t - Y_{t-1} \\ 13 - 8 &= 5 \\ 19 - 13 &= 6 \\ 26 - 19 &= 7 \\ 34 - 26 &= 8\end{aligned}$$

Second difference

$$\Delta^2 Y_t = \Delta Y_t - \Delta Y_{t-1}$$

make them different from previous 15 questions and when you answer the ques, add some detail to explain the terms and concept

Q1. Compute the coefficient of variation for the time series data: 20, 25, 30, 35, 40. Explain its significance.

Coefficient of variation measures relative variability in data and is useful for comparing variation between datasets. First compute mean:

$$\bar{Y} = \frac{20 + 25 + 30 + 35 + 40}{5} = 30$$

Variance

$$\sigma^2 = \frac{(20 - 30)^2 + (25 - 30)^2 + (30 - 30)^2 + (35 - 30)^2 + (40 - 30)^2}{5} = 50$$

Standard deviation

$$\sigma = \sqrt{50} = 7.07$$

Coefficient of variation

$$CV = \frac{\sigma}{\bar{Y}} \times 100 = \frac{7.07}{30} \times 100 = 23.57\%$$

Thus **CV $\approx$ 23.57%**, indicating moderate variability in the dataset relative to its mean.

Q2. Using the linear trend model $Y_t = a + bt$, estimate parameters if data are: Year (t):1,2,3,4 and Sales: 10, 14, 18, 22.

First calculate means:

$$\bar{t} = 2.5, \bar{Y} = 16$$

Slope

$$b = \frac{\sum(t - \bar{t})(Y - \bar{Y})}{\sum(t - \bar{t})^2}$$

$$b = \frac{20}{5} = 4$$

Intercept

$$a = \bar{Y} - b\bar{t} = 16 - 4(2.5) = 6$$

Thus the trend equation is

$$Y_t = 6 + 4t$$

This model shows that sales increase by **4 units each time period**, representing a strong upward trend.

Q3. Compute lag-1 autocovariance for the series: 3, 6, 9, 12, 15.

Mean

$$\bar{Y} = 9$$

Autocovariance formula

$$\gamma_1 = \frac{1}{n} \sum (Y_t - \bar{Y})(Y_{t-1} - \bar{Y})$$

Products:

$$(-3)(-6) = 18$$

$$(0)(-3) = 0$$

$$(3)(0) = 0$$

$$(6)(3) = 18$$

Sum = 36

$$\gamma_1 = \frac{36}{5} = 7.2$$

Positive autocovariance indicates strong persistence where increases tend to follow previous increases.

Q4. Estimate the next value using the AR(2) model $Y_t = 0.5Y_{t-1} + 0.3Y_{t-2}$ for dataset: 10, 12, 15.
Here

$$Y_{t-1} = 15, Y_{t-2} = 12$$

Substitute values

$$\begin{aligned} Y_t &= 0.5(15) + 0.3(12) \\ &= 7.5 + 3.6 \\ &= 11.1 \end{aligned}$$

Thus predicted value is **11.1**. AR(2) models capture influence from the previous two observations, making them more flexible than AR(1).

Q5. Apply weighted moving average with weights 0.5, 0.3, 0.2 for the data: 40, 45, 50.

Weighted moving average gives more importance to recent data.

Formula

$$F = 0.5Y_t + 0.3Y_{t-1} + 0.2Y_{t-2}$$

Substitute values

$$\begin{aligned} F &= 0.5(50) + 0.3(45) + 0.2(40) \\ &= 25 + 13.5 + 8 \\ &= 46.5 \end{aligned}$$

Thus the **forecast value = 46.5**, emphasizing the most recent observation.

Q6. Compute Mean Absolute Error (MAE) for actual values: 100, 110, 120, 130 and forecasts: 98, 115, 118, 132.

Error formula

$$e_t = Y_t - \hat{Y}_t$$

Absolute errors:

2, 5, 2, 2

MAE formula

$$\begin{aligned} MAE &= \frac{\sum |e_t|}{n} \\ &= \frac{2 + 5 + 2 + 2}{4} \\ &= 2.75 \end{aligned}$$

Thus **MAE = 2.75**, which measures the average magnitude of forecasting errors.

Q7. Calculate seasonal index if quarterly sales are: Q1=200, Q2=240, Q3=260, Q4=300 and average annual sales = 250.

Seasonal index formula

$$SI = \frac{\text{Actual value}}{\text{Average}} \times 100$$

Q1

$$= 200/250 \times 100 = 80$$

Q2

$$= 96$$

Q3

$$= 104$$

Q4

$$= 120$$

These indices indicate that **Q4 has the highest seasonal demand**, while Q1 is the lowest.

Q8. Determine if the series shows a linear trend using first differences: 5, 9, 13, 17, 21.

Compute differences

$$\begin{aligned} 9 - 5 &= 4 \\ 13 - 9 &= 4 \\ 17 - 13 &= 4 \\ 21 - 17 &= 4 \end{aligned}$$

Since all differences are constant, the data follow a **linear trend**. Such data can be modeled using

$$Y_t = a + bt$$

where b represents constant growth.

Q9. Compute covariance matrix for two variables $X = (2,4,6)$ and $Y = (1,3,5)$.

Means

$$\bar{X} = 4, \bar{Y} = 3$$

Covariance

$$\begin{aligned} Cov(X, Y) &= \frac{(2-4)(1-3) + (4-4)(3-3) + (6-4)(5-3)}{3} \\ &= \frac{4 + 0 + 4}{3} = 2.67 \end{aligned}$$

Thus covariance matrix

$$\begin{bmatrix} Var(X) & Cov(X, Y) \\ Cov(Y, X) & Var(Y) \end{bmatrix}$$

shows relationships between variables in multivariate time series analysis.

Q10. Forecast next value using exponential smoothing with $\alpha = 0.4$, $Y_t = 80$, previous forecast $F_t = 75$.

Formula

$$\begin{aligned} F_{t+1} &= \alpha Y_t + (1 - \alpha) F_t \\ F_{t+1} &= 0.4(80) + 0.6(75) \\ &= 32 + 45 \\ &= 77 \end{aligned}$$

Thus next forecast = **77**. Exponential smoothing adapts quickly to recent changes.

Q11. Explain the concept of stationarity and test it using dataset: 2,4,6,8,10.

Stationarity means the statistical properties such as mean and variance remain constant over time. A stationary series satisfies

$$\begin{aligned} E(Y_t) &= \mu \\ Var(Y_t) &= \sigma^2 \end{aligned}$$

For dataset mean = 6 but values continuously increase, indicating a **trend**. Thus the series is **non-stationary**. To make it stationary we apply differencing

$$\Delta Y_t = Y_t - Y_{t-1}$$

which produces constant values (2,2,2,2), removing the trend.

Q12. Explain the concept of heteroscedasticity with example dataset: 2, 4, 8, 16, 32.

Heteroscedasticity occurs when variance changes over time. Variance is expressed as

$$Var(Y_t) = \sigma_t^2$$

In this dataset, the magnitude of values grows rapidly, meaning variability increases over time. Such data often appear in financial markets where volatility changes. To stabilize variance, transformations like

$$Z_t = \log(Y_t)$$

or Box-Cox transformation are applied before modeling.

Q13. Explain the Partial Autocorrelation Function (PACF) using dataset: 5, 7, 9, 11, 13.

PACF measures correlation between Y_t and Y_{t-k} after removing intermediate effects. It helps identify the order of AR models. For example, strong correlation at lag-1 but weak correlation afterward suggests an **AR(1)** model. Mathematically the PACF isolates direct relationships between observations separated by k periods.

Q14. Explain the concept of Vector Autoregressive (VAR) models using two datasets: GDP = (100,110,120), Investment = (40,45,50).

VAR models analyze relationships among multiple time series simultaneously. Each variable depends on its own past values and past values of other variables. A VAR(1) model is

$$Y_t = A_1 Y_{t-1} + \varepsilon_t$$

where Y_t is a vector containing GDP and Investment. Such models help study economic interactions and policy effects over time.

Q15. Explain Principal Component Analysis (PCA) in time series forecasting using dataset variables X1, X2, X3.

PCA reduces dimensionality of correlated variables by transforming them into a smaller set of independent components. The transformation is

$$Z = WX$$

where W contains eigenvectors of the covariance matrix. The first principal component explains the maximum variance. PCA is useful in time series forecasting when dealing with many correlated economic indicators because it simplifies models and improves prediction accuracy.